Mallipudi Swarnika

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PROFESSIONAL SUMMARY

AI/ML-focused Computer Science graduate with hands-on experience in deep learning, computer vision, NLP, and full-stack AI development. Skilled in building and deploying intelligent solutions using modern AI technologies, with a focus on healthcare, accessibility, and biometric security applications.

EDUCATION

GMR Institute of Technology — Andhra Pradesh, India
B.Tech -- Computer Science and Engineering (Artificial Intelligence and Machine Learning)
2022 – 2026

Sri Chaitanya Junior College — Andhra Pradesh, India
Intermediate
2020 – 2022

DAV Centenary Public School — Andhra Pradesh, India
CBSE – 86.6% — 2019 – 2020

SKILLS

Programming Languages: Python, Java, SQL, TypeScript, Javascript

AI/Machine Learning: Deep Learning, Computer Vision, NLP, Reinforcement Learning, Linear & Logistic Regression, CNN architectures (MobileNet, FaceNet), Predictive Modelling

Libraries & Frameworks: PyTorch, TensorFlow, Scikit-learn, OpenCV, MediaPipe, NumPy, Pandas

Backend & Databases: Pinecone (Vector Database), PostgreSQL, Prisma ORM, Node.js, Express.js, Flask

Tools & MLOps: Git, macOS Terminal environment, Docker, Jupyter Notebook, Google Colab, Tesseract OCR

EXPERIENCE (INTERNSHIP)

Lomaa IT Solutions — *Dec 2025 – Mar 2025*
Artificial Intelligence Intern

- *Completed a full-semester offline internship focused entirely on Artificial Intelligence .*
- *Developed a Hybrid Face Recognition System capable of real-time face detection and recognition using deep learning and traditional image processing techniques.*
- *Implemented PyTorch and FaceNet for high-dimensional facial feature extraction, normalizing embeddings for improved matching accuracy.*
- *Integrated Pinecone vector database for the efficient storage and rapid retrieval of facial embeddings, reducing computational latency during real-time processing.*

Agratas EduTech — *June 2025 – Aug 2025*
Machine Learning Intern

- Developed and optimized machine learning models using Python, Scikit-learn, Pandas, and NumPy.
- Performed data preprocessing, feature engineering, and parameter tuning for supervised and unsupervised algorithms on real-world datasets .

Prodigy InfoTech — *Feb. 2025*
Machine Learning Intern

- *Developed and deployed 5+ machine learning and deep learning models for automation, safety, and data analysis applications.*

MSME Technology Centre, Visakhapatnam — *Duration: 1 Month*
JAVA INTERN

- *Gained foundational knowledge in Java and software development practices.*

KEY AI PROJECTS

AI-Enabled Personalized Medication Tracker

Technologies: Python, TensorFlow (MobileNet), Tesseract OCR, Reinforcement Learning, NLP, Next.js, Node.js, PostgreSQL, Prisma ORM, Docker

- Engineered a full-stack medication tracking ecosystem.
- Built a visual pill classifier using a MobileNet CNN architecture that achieved 94.8% classification accuracy, alongside a Tesseract OCR module that reached 94.05% accuracy for text extraction from medicine labels.
- Designed a Reinforcement Learning (RL) smart scheduler that dynamically optimized reminder timings based on patient behavior, improving overall adherence rates.
- Integrated a conversational healthcare chatbot leveraging NLP and BERT/GPT-based models, achieving 93% intent detection accuracy for dosage and side-effect queries.
- Deployed the system using Docker containers and RESTful APIs to ensure low-latency performance and secure data flow.

Real-Time Sign Language to Text and Speech System (SLTS-CNN)

Technologies: Python, CNN, MediaPipe, OpenCV, Google TTS, pyttsx3

- Developed an AI-powered accessibility tool to translate American Sign Language (ASL) gestures into text and audio
- Created a robust CNN pipeline integrated with MediaPipe for precise hand segmentation and OpenCV for real-time video stream processing.
- Achieved an overall gesture recognition accuracy of 91.11% on a custom dataset encompassing alphabets, digits, and punctuation marks.
- Implemented multi-modal output by passing text predictions through Google TTS engines to synthesize real-time speech.

CERTIFICATIONS & PUBLICATIONS

Generative AI Engineer – L&T EduTech: Gained hands-on knowledge of cloud-based generative models, language model development, and computer vision.

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Publication : "A SOFTWARE MOBILE APPLICATION-GROUND WATER LEVEL PREDICATOR." Published in the International Journal of Research Publication and Reviews (IJRPR), Vol. 05, Issue 12, Dec 2024 (Impact Factor: 6.844)